

MXLIMS

for multi-sweep, multi-crystal experiments

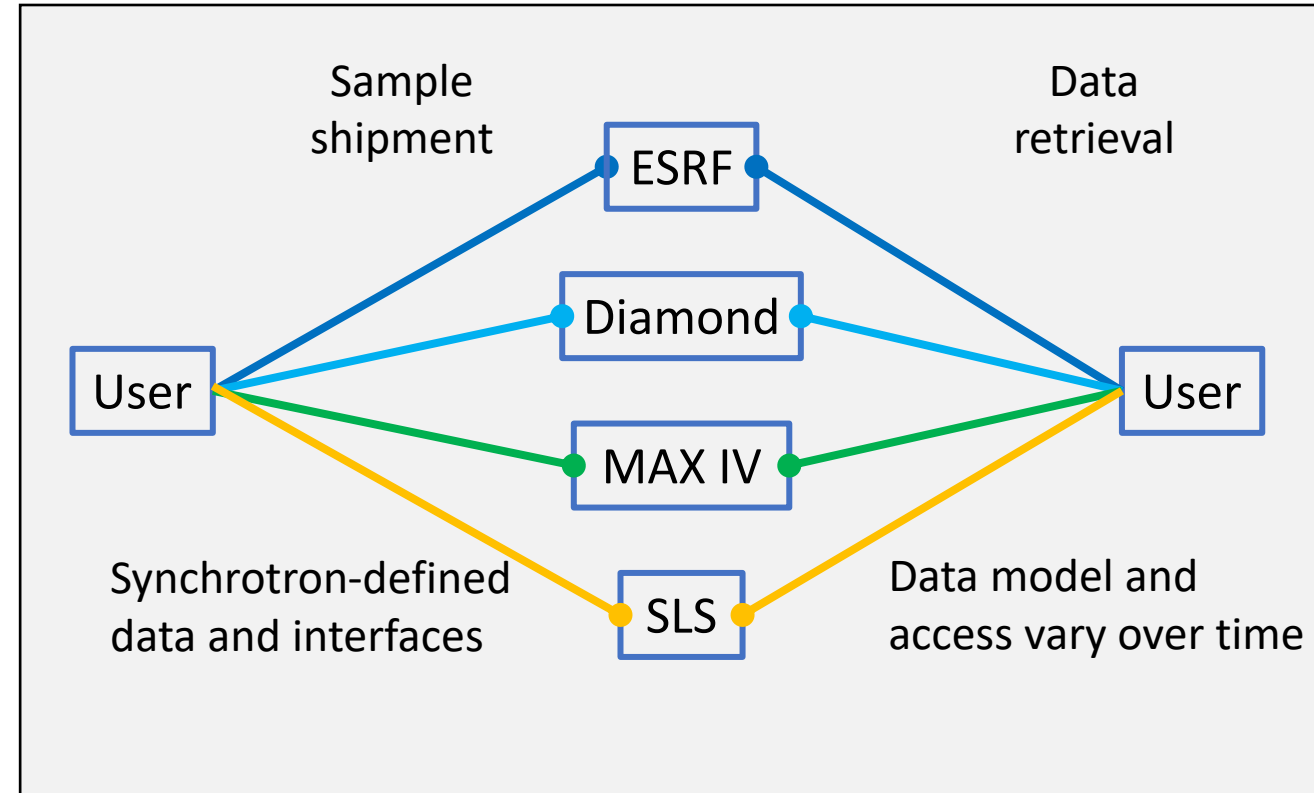
Rasmus Fogh, Global Phasing

MXCuBE/ISPyB Meeting, HZB Berlin, 11 June 2026

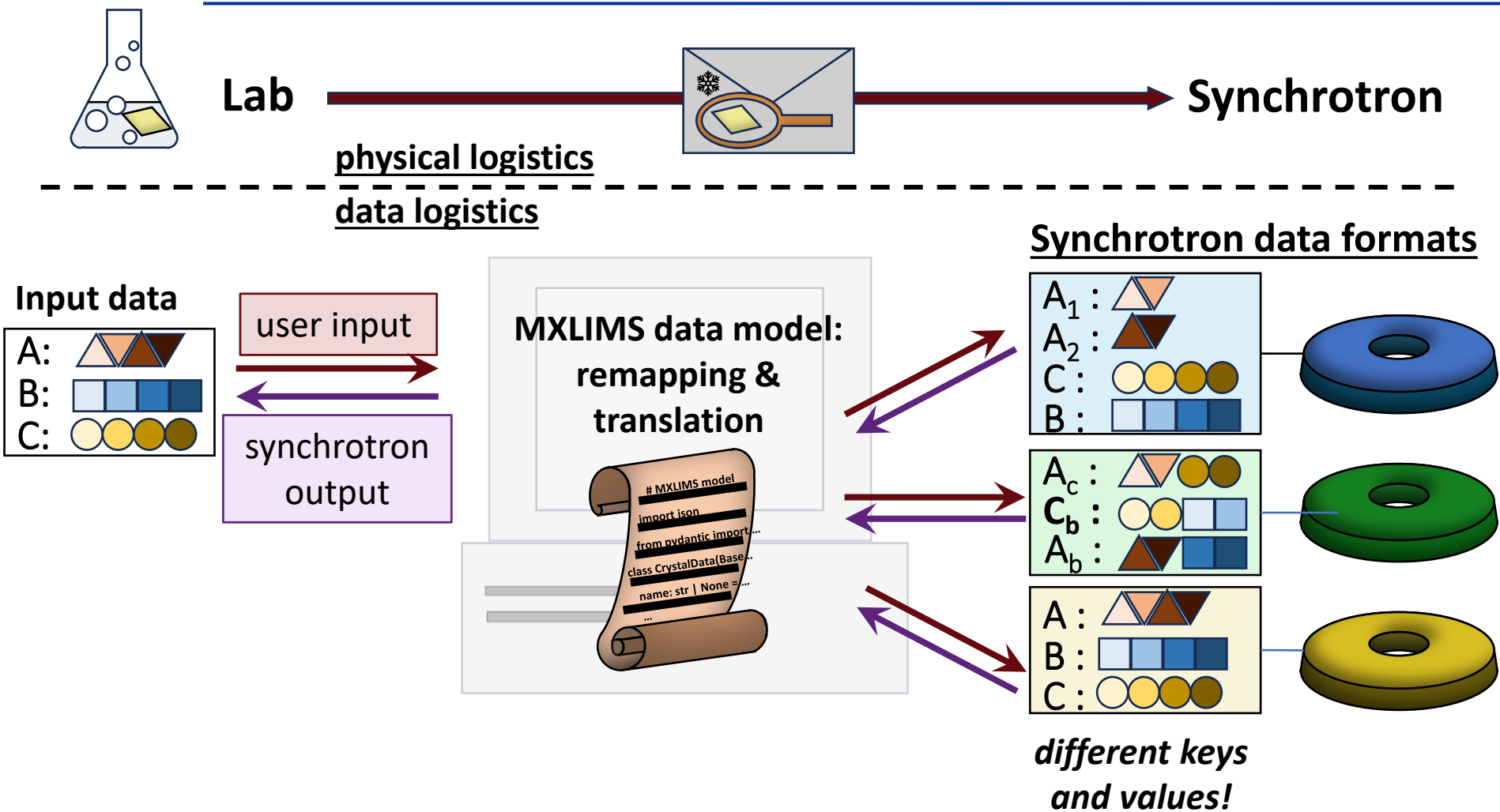
- **MXLIMS Overview**
- Multi-sweep, multi-crystal facilities

Shipping and data retrieval - now

- Separate proposals for each synchrotron
 - Different values, tag names, ...
- Web site access, no software interface
 - Ad-hoc, fragile software connection
- Synchrotrons (re)define formats but users must update data I/O code
 - $N \text{ users} \times M \text{ synchrotrons}$
= $N \times M$ converters to maintain
- Diverse access control procedures and information requirements
 - Access controls will never be the same, but procedures might be harmonised



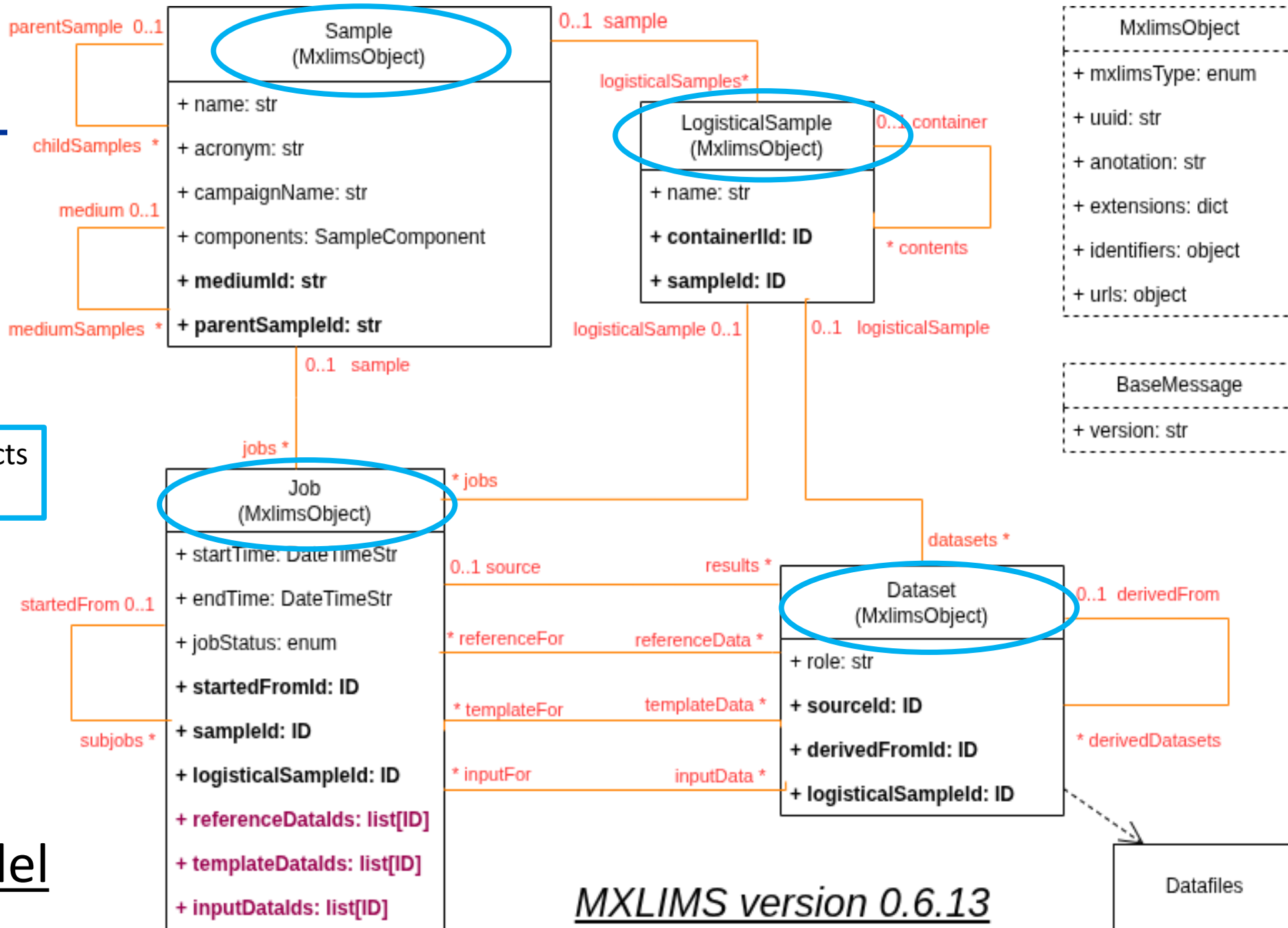
Alternative - using MXLIMS



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- A clearly defined standard using JSON schemas
 - JSON message files validated against the standard
 - Code generation for implementations (Pydantic, potentially noSQL databases)
 - Flexibility and maintainability (*unlike ISPyB* 😊)
 - Slots for site-specific fields and identifiers, to allow extension
 - Data file versioning, so you can support multiple versions in parallel
 - A simple domain-independent core
 - Metadata repository
 - Support for provenance tracking
 - Independent extension to new domains
 - Reuse schemas for multiple, similar objects (e.g. diffraction plan, results)
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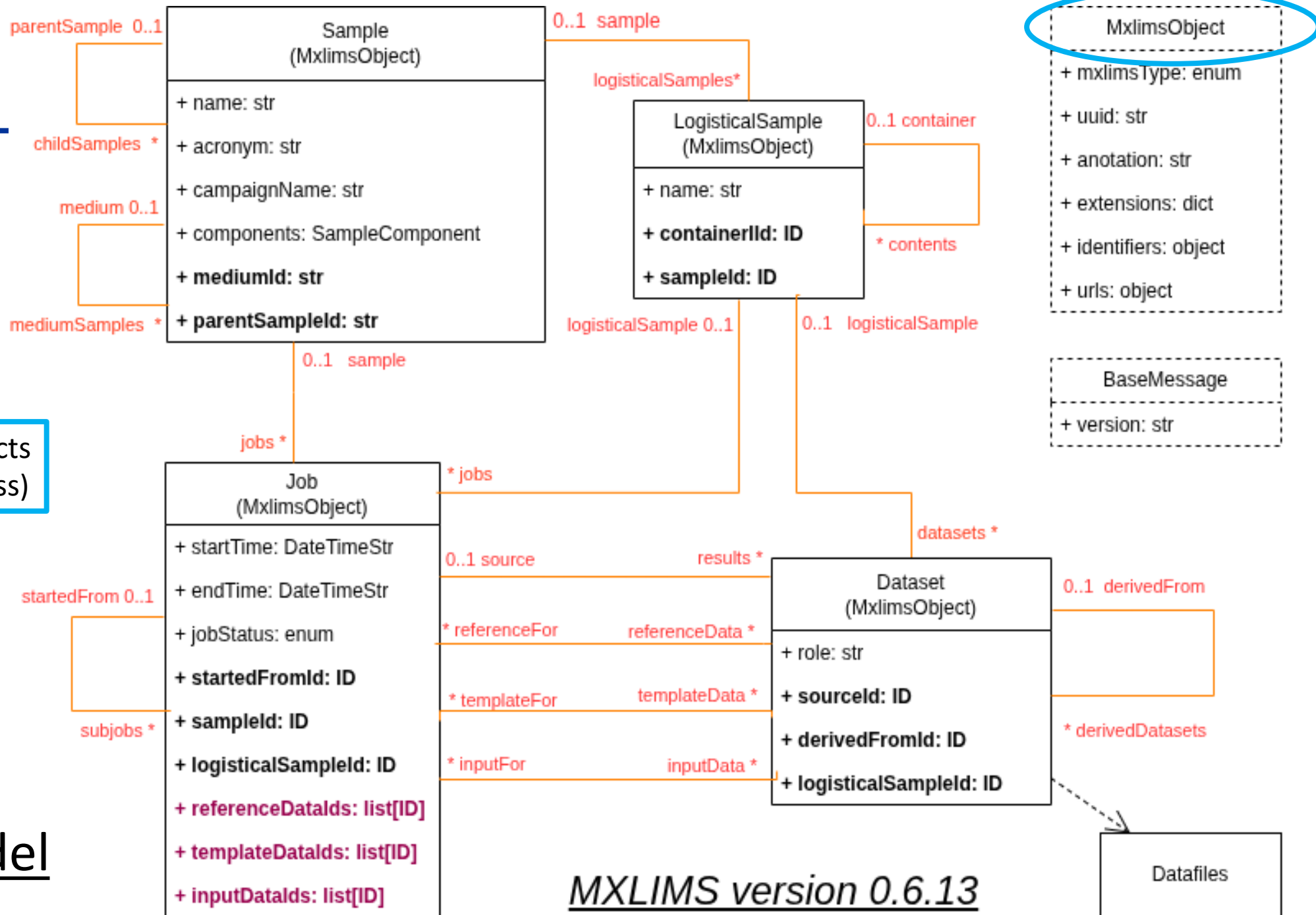
Four kinds of objects
(‘noSQL tables’)

Core Model

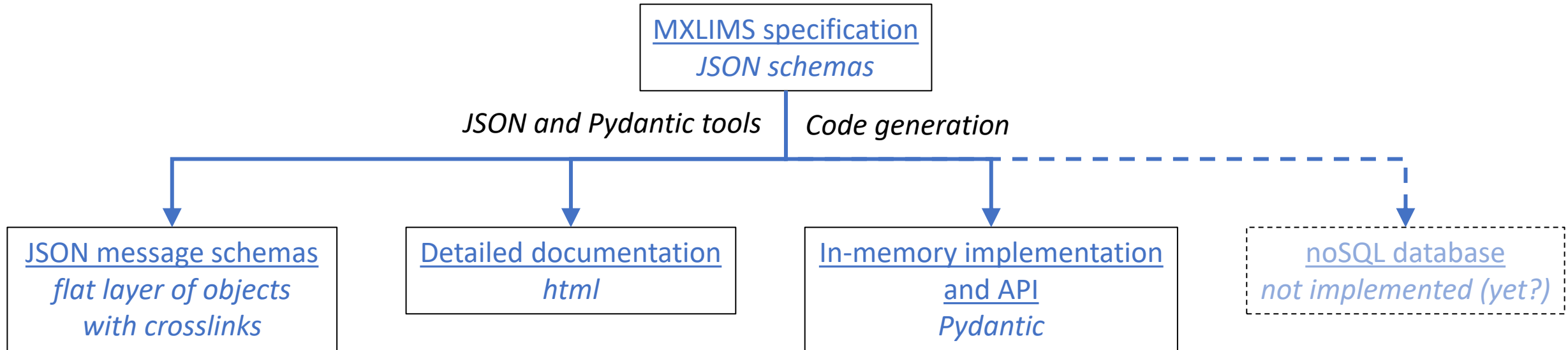


Shared by all objects
(abstract superclass)

Core Model



MXLIMS implementation

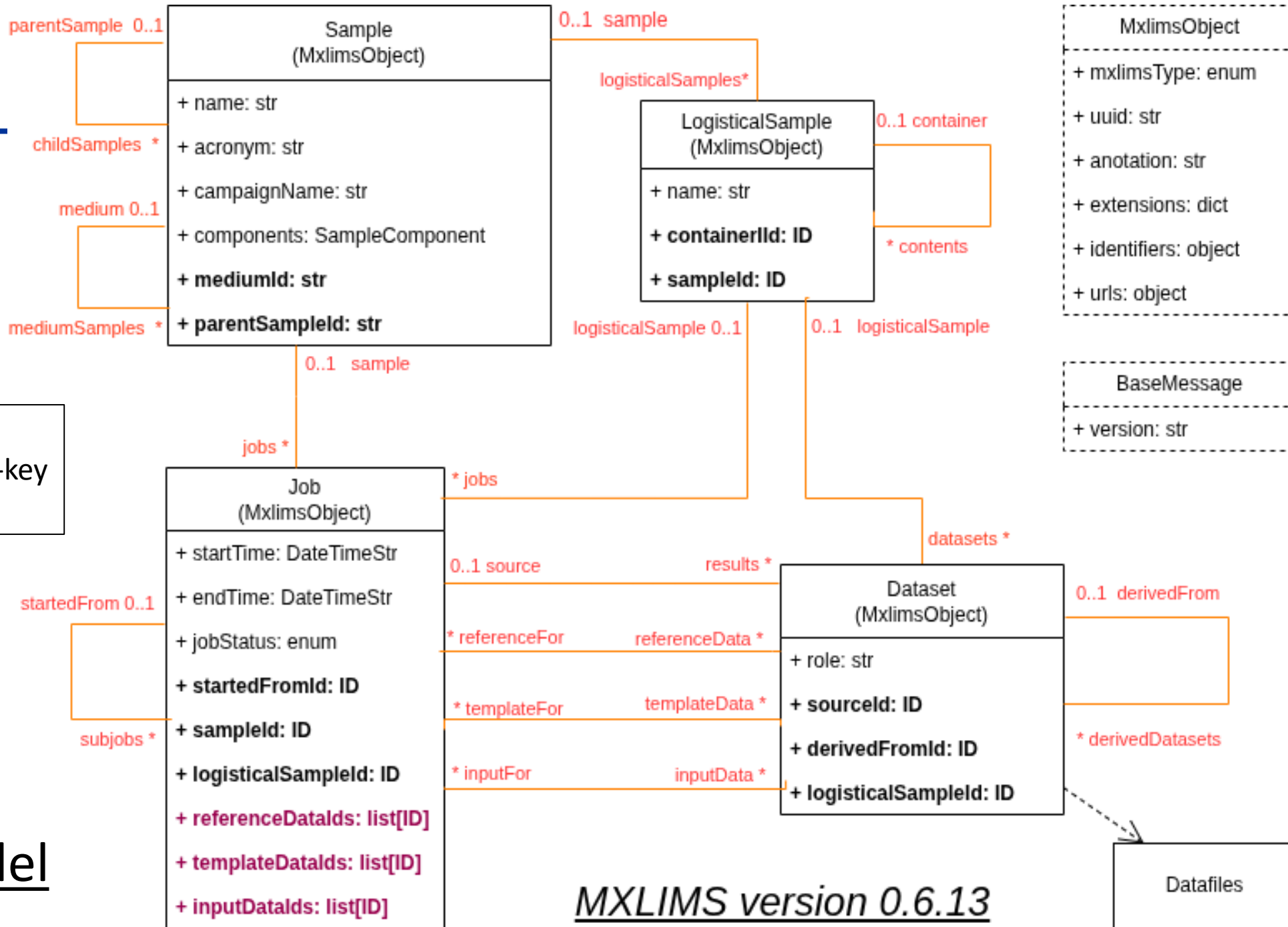


JSON supports tree structures, but objects form a web, not a tree.

Links between objects are stored as object ID pointers (foreign keys, UUIDs) also within messages.

Link handling and link type checking differs between messages, API, and different implementations.

Code generation is used to adapt the basic specification to each case.



Model data - examples

MxExperiment (Job)

- + experimentStrategy: str
- + expectedResolution: float
- + expectedUnitCell: UnitCell
- + priority: int
- + radiationSensitivity: float
- + selectedUnitCell: UnitCell

MxProcessing (Job)

- + programName: str
- + programVersion: str

CollectionSweep (Dataset)

- + sweepType: str
- + energy: float
- + exposureTime: float
- + imageWidth: float
- + meshRange: [float, float]
- + numberImages: int
- + path: str
- + fileNameTemplate: str

ReflectionSet (Dataset)

- + wavelengths: List[float]
- + anisotropicDiffraction: bool
- + binningMode: str
- + numberBins: int
- + path: str
- + fileName: str
- + spaceGroupName: str
- + refinedUnitCell: UnitCell
- + operationalResolution: float
- + diffractionLimitsEstimated: Tensor
- + reflectionStatisticsOverall: ReflectionStatistics
- + reflectionStatisticsShells: List[ReflectionStatistics]

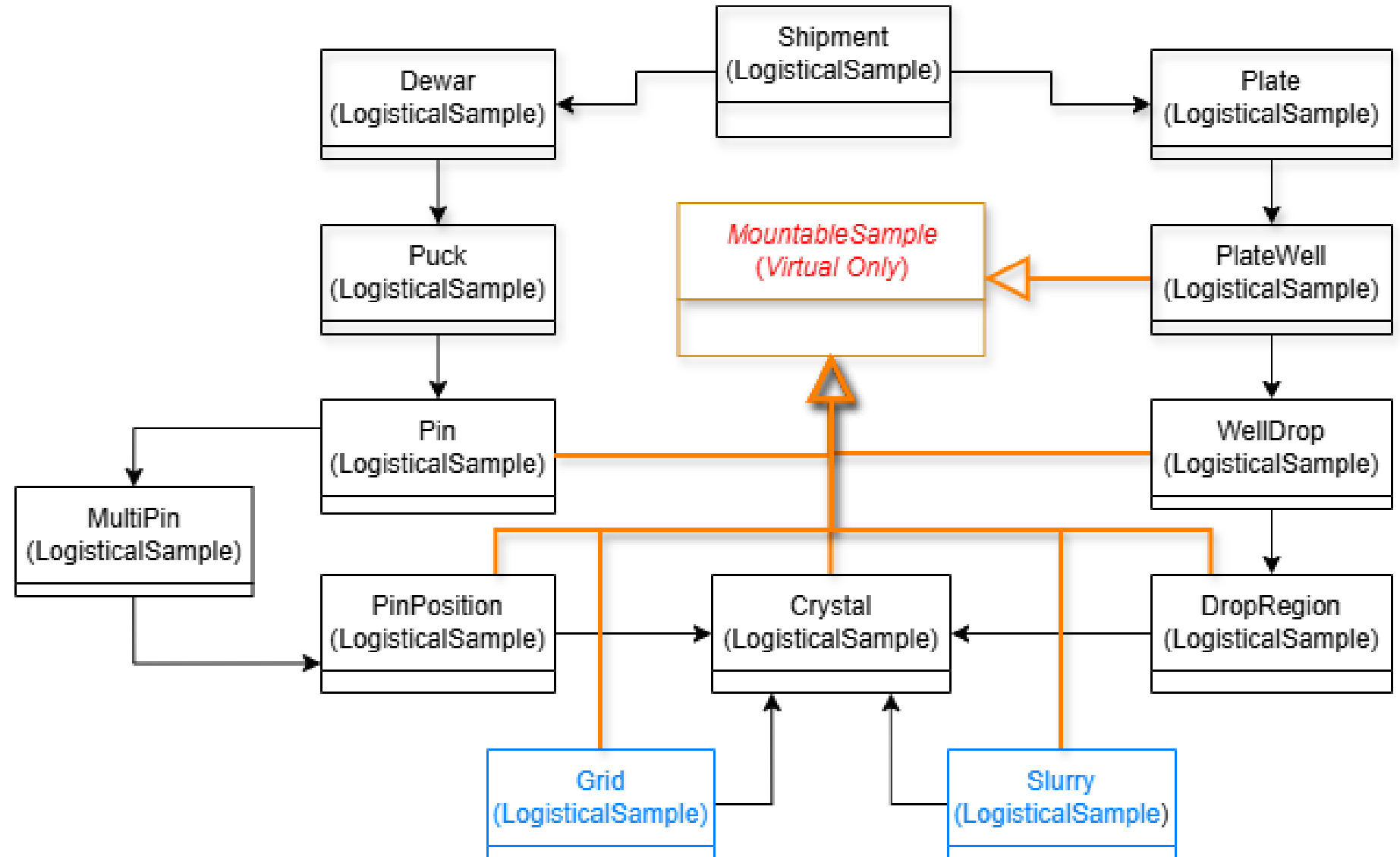
MacromoleculeSample (Sample)

- + name: str
- + acronym: str
- + campaignName: str
- + components: List[SampleComponent]

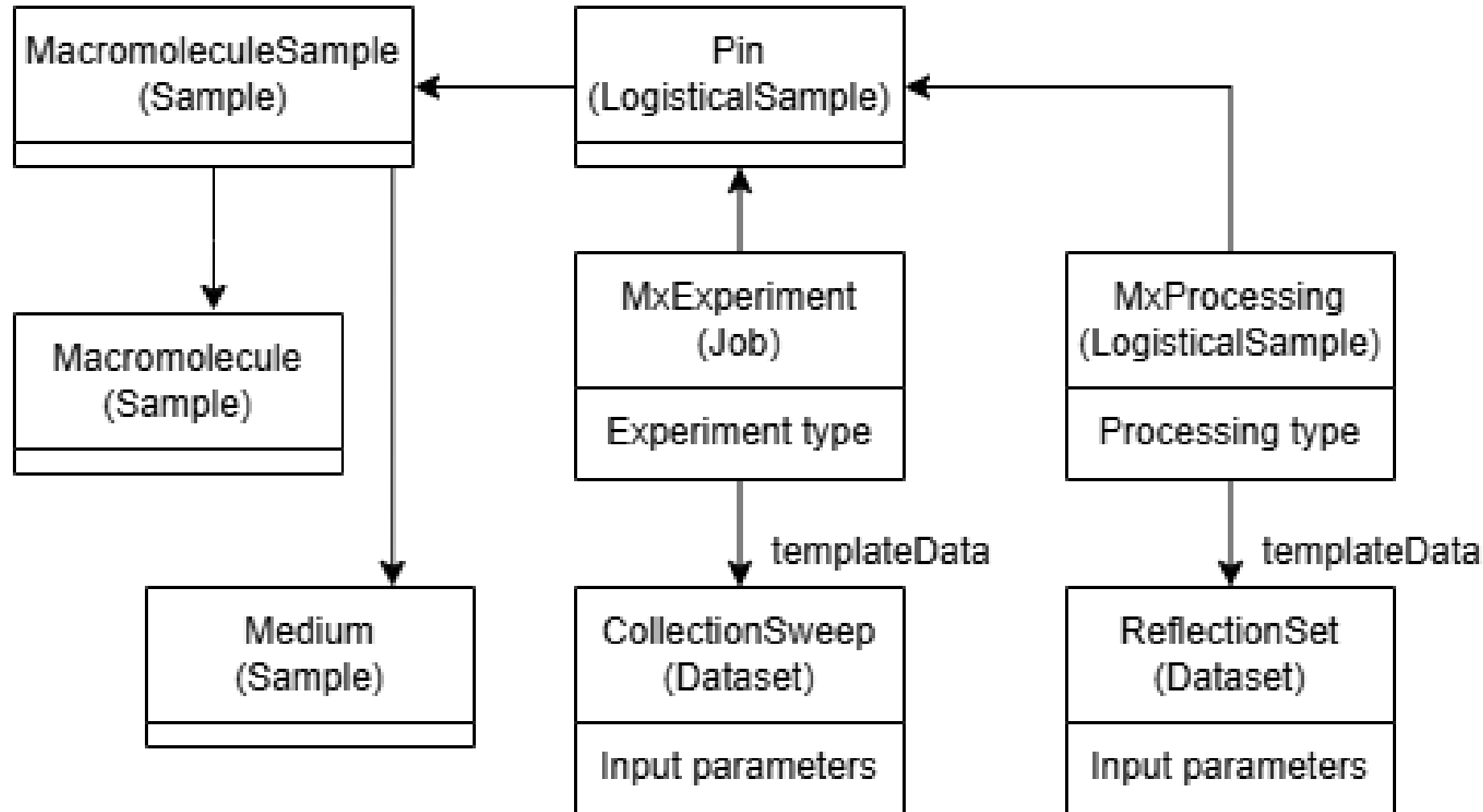
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- ‘Mountable samples’ (not a real class) are samples that can be used to run an experiment on.
- Grids and slurries are not currently modeled, but would fit in as samples for SSX or XFEL.
- Crystal objects are only generated as needed. In grids, slurries, multi-crystal wells, or lipid cubic phase, crystals cannot be identified before the experiment

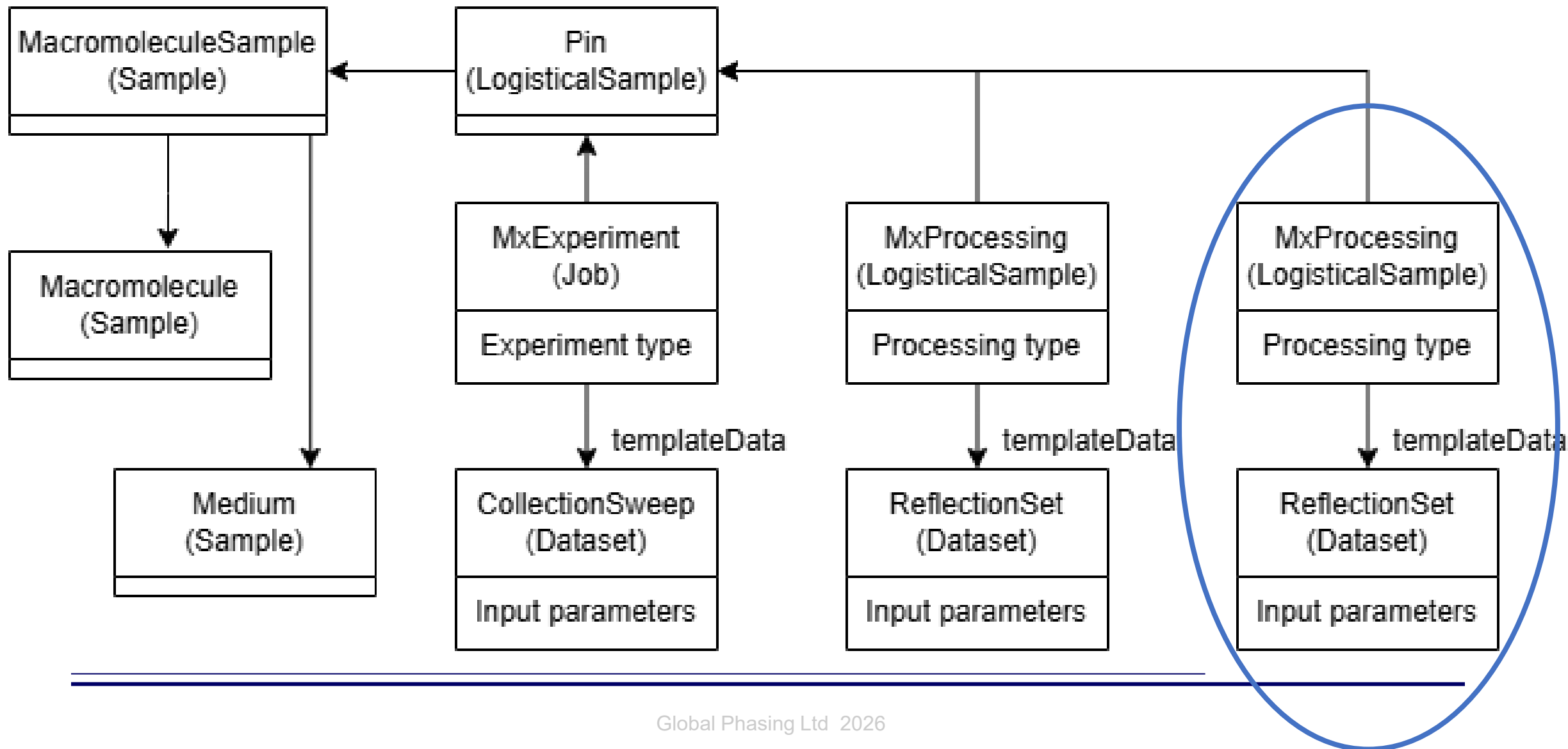
Logistical Samples



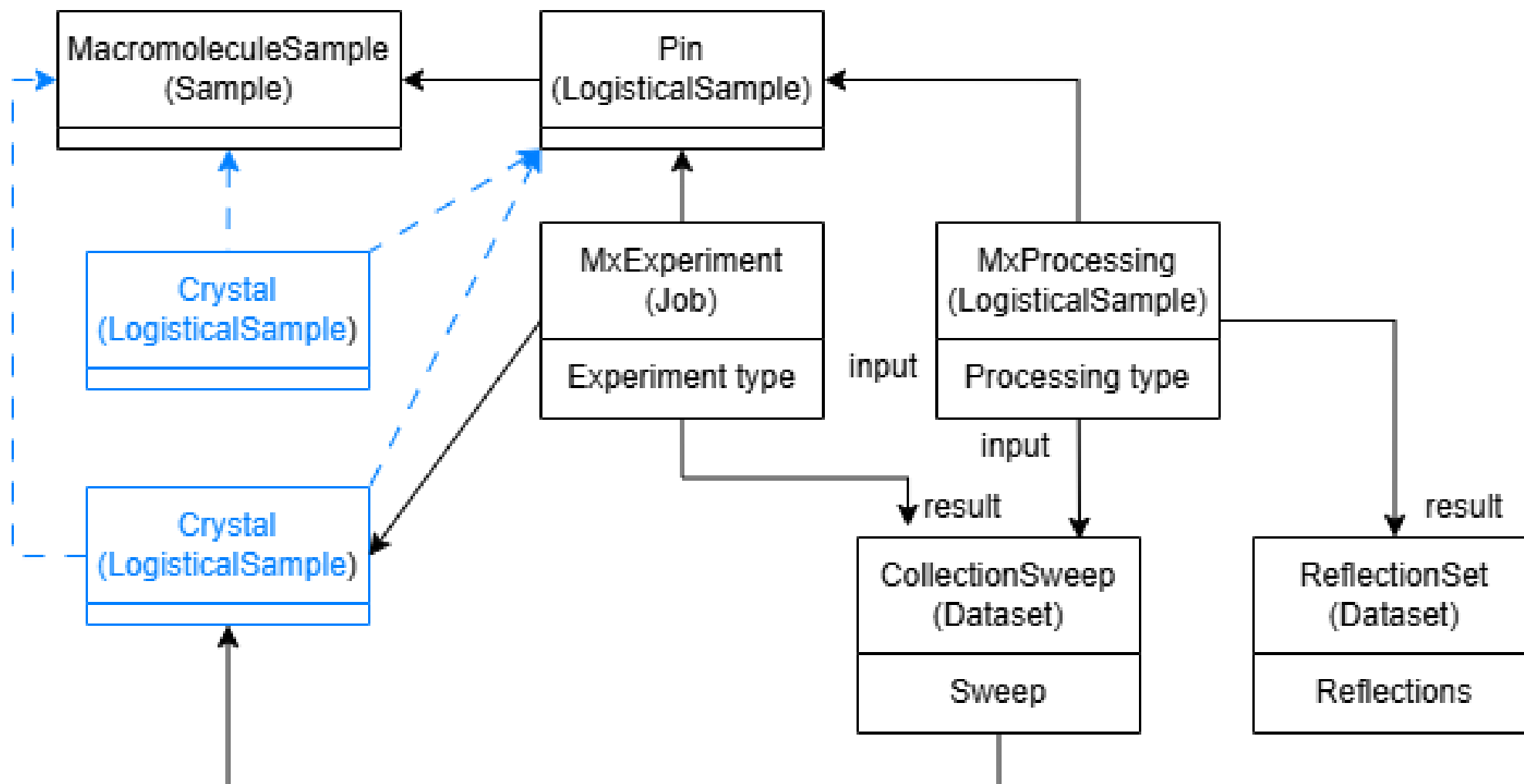
Experiment input



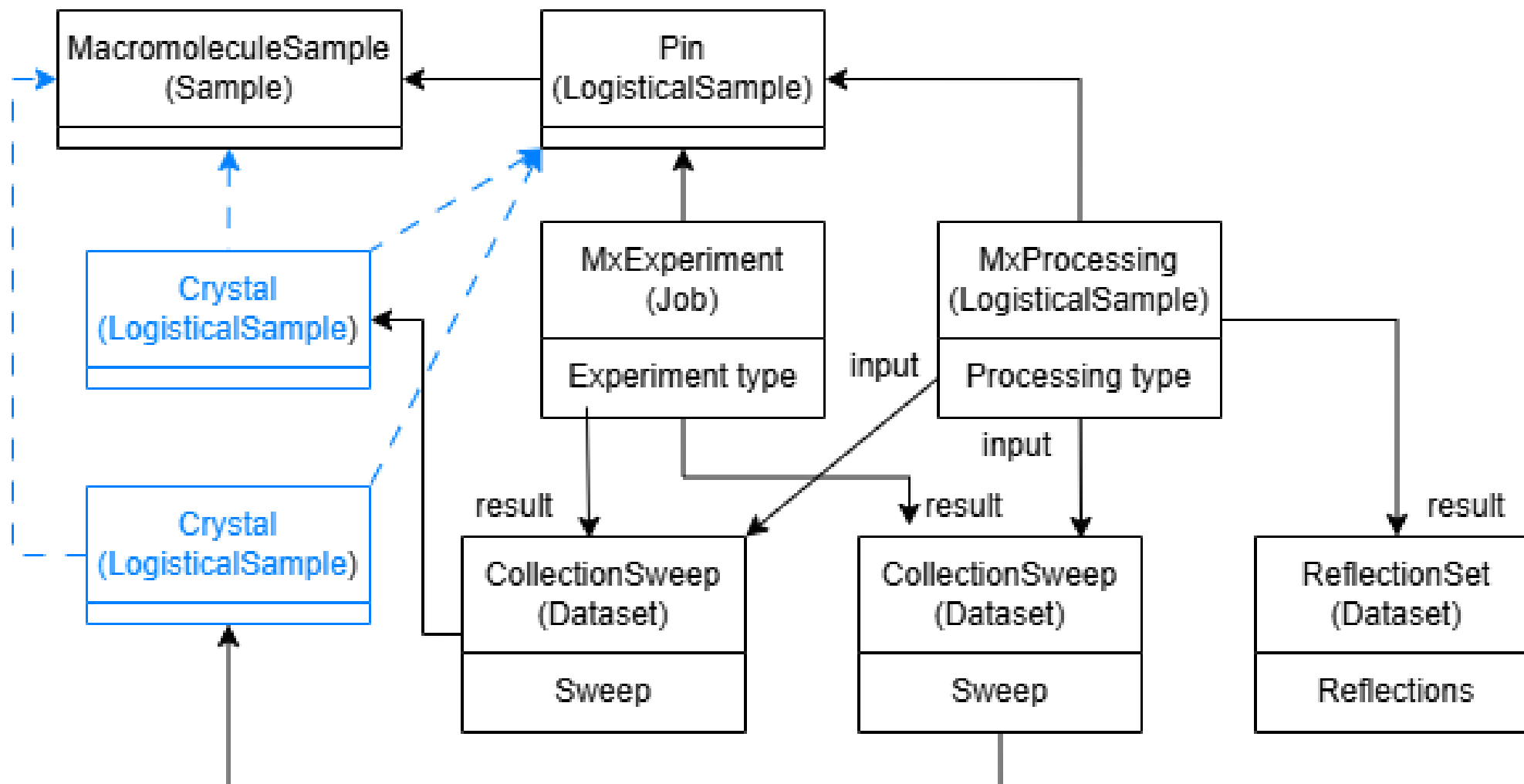
Experiment input - multiple processing



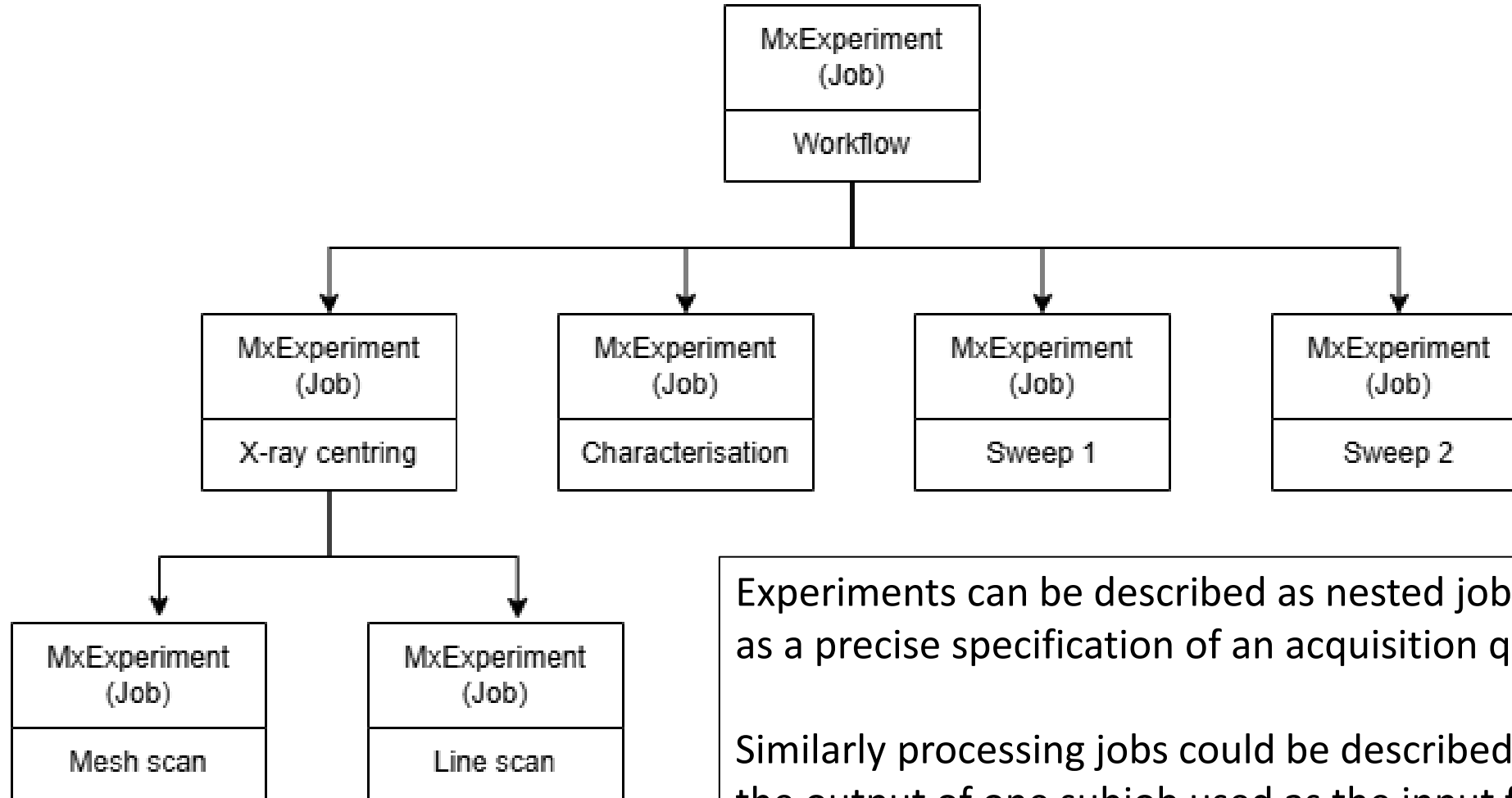
Experiment results



Multi-crystal multi-sweep experiment



Nested jobs / subjobs



Experiments can be described as nested jobs / subjobs, e.g. as a precise specification of an acquisition queue.

Similarly processing jobs could be described as nested, with the output of one subjob used as the input for others.

- MXLIMS is not a private project, but a nascent community standard
- As new participants join, the project will be reorganised:
 - Shared, public repositories
 - Procedures for accepting changes, making new versions etc.
 - Governance by stakeholders
- MXLIMS is ready for you to join in!

- Thanks to the many, many people who have participated in the MXLIMS working group, the MXCuBE automation WG, or contributed information, discussions and feedback.
 - For work on integration, synchrotron mapping and testing and making the MXLIMS model, particular thanks to Guilherme de Freitas, James Hall, Jose Brandao-Neto and Karl Levik (Diamond), Daniele de Sanctis, Didier Nurizzo, Marcus Oscarsson, Olof Svensson, and Yan Walesch (ESRF), Leigh Carter, Clemens Vonnrhein and Peter Keller (Global Phasing), Ed Daniel (IceBear), Ezequiel Panepucci (MAX IV), Dennis Stegmann and May Sharpe (SLS), Martin Savko (SOLEIL), and Gustavo Lima (Astex)
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